

Define Problem Statements

HDI Prediction System | Team Lead: Farzana | Contributors: Chinthoti Raghavendra, Ahmad Kabir Shaik, Naga Sireesha Naralasetty, Sitha Ramaraju Velisala

Problem Statement

Government bodies and researchers need the Human Development Index (HDI) to compare countries and guide policy decisions. However, official HDI scores are published only once a year, and calculating HDI from raw indicators is a manual, technical process, leading to delayed insights and limited accessibility for non-experts.

Who Is Affected

- Policy researchers who need up-to-date HDI estimates for time-sensitive decisions.
- Students and analysts who lack the statistical background to compute HDI manually.
- Government bodies that bear the cost and delay of yearly, centralized HDI reporting.

Problem Statement (Refined)

"How might we help researchers and policymakers estimate a country's HDI quickly and accurately using available socio-economic indicators, so that development analysis is faster and more accessible?"

Goals

- Predict HDI instantly from key indicators instead of waiting for annual reports.
- Provide a reliable, data-driven estimate using a trained regression model ($R^2 \approx 0.98$).
- Make the tool accessible through a simple web interface, needing no statistical expertise.

Non-Goals (Out of Scope for MVP)

- Replacing official UNDP HDI reporting or methodology.
- Real-time integration with live government or census databases.
- Country-level policy recommendations beyond the predicted HDI score (academic project only).